LUMEN

LUNAR LANDING SITE NAVIGATOR

PROJECT BRIEF · PRODUCT WORKING GUIDE · TECHNICAL DETAILS

Know when the light goes out — before you land.

A guided, data-transparent explorer for lunar surface illumination and geometric direct-to-Earth visibility.

Project snapshot

NASA Space Apps Challenge 2026 · Cairo local event | Team lead: Amr Alshabasy | Document snapshot: 13 November 2026

Purpose of this document

This guide explains what LUMEN does, how a user works through it, where its numbers and visuals come from, how to run the project, and what its current validation does and does not establish.

Implementation facts below are based on the project playbook, checked source files, configuration, generated datasets, and validation notes in this workspace. Live ephemeris availability depends on NASA/JPL Horizons and the deployed API route.

01 · OVERVIEW

What LUMEN is

LUMEN helps people explore when the Sun and Earth appear above a selected lunar site's local horizon over a defined time window. It brings a cited site, a shared time cursor, a sky view, an interactive 3D scene, comparison metrics, and method notes into one visual workflow.

The key planning question is simple: if a landing date or candidate site changes, how do the geometric illumination and Earth visibility windows change? LUMEN makes that screening comparison easier to inspect without hiding the source, frame, time system, or limitations.

Scope and interpretation

LUMEN is a geometric screening and educational visualization. It is not NASA guidance, landing approval, a communications link guarantee, a power/thermal model, or live spacecraft telemetry. The local horizon is a flat reference sphere; terrain is not modeled.

Who it serves

Audience	How LUMEN helps
Mission planners and analysts	Quickly compare site/time geometry and inspect provenance before deeper terrain or link analysis.
Educators and students	See how local horizon geometry and changing dates affect the Sun and Earth view.
Public and judges	Follow a guided visual story, then inspect the actual assumptions behind the displayed values.

Working principle

The reproducible site timeline is calculated offline from a pinned NAIF SPICE kernel set and shipped as static JSON. A separate current-epoch panel and the orbit-context view query NASA/JPL Horizons through same-origin API routes. These two time sources remain visibly distinct.

02 · USER JOURNEY

How a user works through LUMEN

1. Enter through the cinematic landing page. The muted lunar animation, deep-space visuals, and scroll-led sections introduce the planning problem. Media and effects are illustrative; they do not supply scientific results.
2. Move into the celestial viewer. Choose SITE VIEW for the selected lunar surface location or LIVE ORBITS for current Earth–Sun and Moon–Earth orbital context.
3. Select a candidate site. The current catalog contains two Apollo reference sites and three south-pole candidates. The app shows signed latitude/longitude, coordinate frame, and source context.
4. Choose the geometry time source. TIME CURSOR follows the selected precomputed SPICE sample. LIVE NOW follows the current Horizons snapshot at its reported UTC minute.

5. Use the explorer timeline. Scrub or play the shared cursor to update Sun and Earth elevation, horizon status, sky-dome placement, 3D directions, and the next available refined transition.
6. Read the screening card beside its inputs. GO / CAUTION / NO-GO is derived from visible summary fractions and documented placeholder thresholds; it is not a recommendation.
7. Enable compare mode. Select Site A and Site B, then read the shared-cursor values and Site B minus Site A deltas. The app warns if the generated windows or steps do not match.
8. Open method, source, and limit notes. Review coordinate provenance, frames, kernels, Horizons validation, and the persistent terrain limitation before interpreting a result.

Main controls and interactions

Control	What it changes
Site selector	Loads one of the five curated, cited site datasets.
Time cursor / play	Moves through the static 15-minute samples; shared by metrics and visualization.
SITE VIEW / LIVE ORBITS	Switches between local surface geometry and current JPL state-vector context.
TIME CURSOR / LIVE NOW	Chooses which epoch source drives Sun/Earth directions in SITE VIEW.
Mouse / touch in 3D	Drag to rotate, wheel/pinch to zoom, right-drag/two-finger pan to travel; reset restores framing.
Compare toggle and site selector	Displays two sites on the same cursor/window and computes signed differences.
Audio and motion controls	Optional synthesized sound after a user gesture; reduced-motion preferences are respected.

03 · FEATURES

Feature inventory

Feature	Current behavior	Data / caveat
Cinematic landing page	Muted, automatically played lunar video; scroll narrative; section reveals and navigation.	Video is a visual asset only; playback can be subject to browser policy.
Candidate-site explorer	Five sites with coordinates, source links, status metrics, and selected site marker.	Cited coordinates in MOON_ME_DE440_ME421; planetocentric latitude, east-positive longitude.
Local sky and horizon	Sun/Earth azimuth and elevation, above/below status, Earth angular size, and local sky representation.	Static cursor samples use SPICE; flat horizon; no terrain masking.

Feature	Current behavior	Data / caveat
3D SITE VIEW	Textured Moon with selected-site marker; Sun/Earth directions follow cursor or LIVE NOW snapshot.	LRO WAC, Blue Marble, and SDO/AIA textures are visual references; positions/dimensions are stylized.
3D LIVE ORBITS	Current Earth heliocentric and Moon geocentric paths, plus their current sampled positions.	Horizons DE441 vectors; real vector paths shown with expanded, non-comparable visual distances.
Time-window metrics	Sunlit fraction, Earth-center-visible fraction, longest dark period and Earth outage where present in dataset.	Derived from the generated sample window and model rules; not an engineering power or link model.
Refined transitions	Displays the next event recorded in the generated dataset and its countdown from the cursor.	Use only events included in the selected static window; no event is invented outside coverage.
Compare mode	Side-by-side values and signed deltas at one shared sample time.	Like-for-like requires matching window, step, and sample count.
Current Horizons snapshot	Sun/Earth apparent azimuth, elevation, angular diameter and UTC snapshot; refreshes every minute.	Ephemeris snapshot, not telemetry; requires network and working API deployment.
Method and transparency	Exposes data sources, frames, limits, validation, visual credits, and placeholder verdict criteria.	Four sites have documented spot checks; Malapert and event crossings need further checks.

Screening indicator

The frontend calculates the indicator from the selected dataset's summary and config/verdict.json. Current placeholder settings: GO requires at least 0.60 sunlit fraction and 0.70 Earth-center-visible fraction; CAUTION boundary values are 0.30 and 0.40. The exact classification logic and raw fractions must be read together. These are project placeholders, not NASA or mission criteria.

Reading the orbit view

The orbit view uses actual current Horizons vectors, but the Earth and Moon distances are expanded substantially so both paths are visible in one scene. Treat the path geometry and epoch as data-backed; treat displayed sizes and relative spacing as illustrative, not to scale.

04 · DATA AND METHOD

Scientific data flow

For the reproducible timeline, config/analysis.json fixes the generated interval to 2027-01-01 00:00 UTC through 2027-03-02 00:00 UTC inclusive, with a 900-second (15-minute) step, LT+S aberration setting, 1737.4 km reference radius, and explicit MOON_ME_DE440_ME421 frame. UTC is used at input/output boundaries; SPICE ET/TDB is used internally. Window summary

fractions cover all generated samples. Horizons documents that UTC conversion after 2027-01-01 holds the last known leap-second offset, so future UTC tags carry that limitation.

Precomputed site geometry

1. Load the documented NAIF kernels in deterministic order and check that the configured interval is covered.
2. Read each site's cited planetocentric latitude and east-positive longitude in the lunar Mean Earth frame. Geometry is evaluated from the lunar surface site, rather than the Moon center.
3. At every time sample, compute site-relative Sun and Earth vectors, transform them into the site East-North-Up basis, and derive azimuth (clockwise from north) and elevation (from the local horizontal).
4. Store time series, apparent Earth angular radius, events, summary fractions, warnings, and provenance in one JSON file per site. Kernel names and hashes are recorded in the generated data provenance.
5. Refine crossings against the continuous geometry with bisection before recording event times; the UI countdown uses only events inside the generated dataset window.

Live API data

The /api/horizons route requests current-minute observer-table data for the Sun and Earth from a curated lunar site and reports apparent azimuth, elevation, and angular diameter. The /api/orbits route requests Horizons vector tables: Earth relative to the Sun across about ±180 days at five-day spacing, and the Moon relative to Earth across about ±15 days at 12-hour spacing. The endpoints use NASA/JPL Horizons and identify DE441 for the orbit vectors.

These routes are proxied through the Vite development server locally and deployed as Vercel serverless functions in production. Both are network-dependent; a failed request should be treated as unavailable live context, not replaced with zero or a fabricated value.

Kernels and coordinates

Input	Configured source / convention
Time and ephemeris	naif0012.tls; de440s.bsp; UTC to ET/TDB through SPICE.
Lunar body orientation	pck00011.tpc; moon_pa_de440_200625.bpc; moon_de440_250416.tf.
Frame	MOON_ME_DE440_ME421, explicitly verified against the loaded DE440 frame kernel.
Local basis	East-North-Up. Azimuth clockwise from north; elevation from local horizontal.
Site coordinates	Planetocentric selenographic latitude; east-positive longitude; 1737.4 km reference sphere.

Candidate sites

All coordinates below are signed values expressed as conventional hemispheres for readability. Exact source and frame metadata are kept in config/sites.json and the generated site files.

Site	Latitude	Longitude E	Role / source basis
Apollo 11 reference	0.67408° N	23.47297° E	Historical reference; NASA Apollo mission site coordinates.
Apollo 12 reference	3.0128° S	23.4219° W	Historical reference; NASA/NSSDC lunar site table.
Shackleton rim	89.7668° S	171.87° W	South-pole candidate; NASA Lunar Surface Data Book / PGDA DEM polygon center.
de Gerlache Rim 1	88.6834° S	Azimuth 0.003° · elevation 0.003°	South-pole candidate; NASA Lunar Surface Data Book / PGDA DEM polygon center.
Malapert Massif	85.9948° S	0.23578° W	South-pole candidate; NASA Lunar Surface Data Book / PGDA DEM polygon center.

The cited south-pole candidate center coordinates are reported by their source as derived from PGDA DEM polygons, with stated center-point accuracy of about 5–10 m. This coordinate provenance does not mean terrain is used in the current horizon calculation.

Visual assets and their meaning

Asset	Source	What it represents
Moon	NASA GSFC SVS / LRO-LROC WAC global mosaic.	Surface appearance map. Polar gaps may be filled using lower-resolution albedo; not terrain data.
Earth	NASA Earth Observatory Blue Marble base map.	Earth appearance reference; not a current or epoch-matched image.
Sun	NASA SDO/AIA 171 Å image (2026-09-01).	False-color ultraviolet solar-atmosphere view; unseen hemisphere mirrored for texture.
Hero video	NASA SVS “The Moon’s Rotation” (720p MP4), credited to NASA SVS and visualization by Ernie Wright; public domain unless the source page identifies an exception.	Cinematic background art. It does not encode current illumination or site geometry.
Stars / effects	Generated visual effects and motion.	Atmosphere only; not a measured star catalog or telemetry display.

Coordinate and imagery references

Reference	Link
Apollo 11 coordinates	https://www.nasa.gov/wp-content/uploads/static/history/SP-4029/Apollo_18-27_LM_Lunar_Landing.htm
Apollo landing site table	https://nssdc.gsfc.nasa.gov/planetary/lunar/lunar_sites.html
NASA Lunar Surface Data Book	https://ntrs.nasa.gov/api/citations/20230007818/downloads/ACD-50044%20Lunar%20Surface%20Data%20Book%20Rev%20A.pdf
NASA GSFC PGDA	https://pgda.gsfc.nasa.gov/products/78
LRO/LROC visual asset	https://svs.gsfc.nasa.gov/4720/
NASA Blue Marble	https://science.nasa.gov/earth/earth-observatory/blue-marble-next-generation/base-map/

06 · LIVE 3D ORBITAL CONTEXT

What the orbit viewer plots

LIVE ORBITS gives spatial context around the Sun–Earth system and the Earth–Moon system. The API obtains two separate vector series from JPL Horizons: heliocentric Earth vectors and geocentric lunar vectors. Their current positions are selected from the closest vector sample to the API epoch.

Property	Value shown by the implementation
Source	NASA/JPL Horizons API, DE441.
Epoch	The response reports epochUtc rounded to the current UTC minute; displayed beside the viewer.
Frame / axes	Ecliptic J2000 reference plane, ICRF axes.
Units	AU in the response.
Correction	Geometric vectors, VEC_CORR=NONE.
Earth series	Target Earth (399), center Sun (500@10), roughly ±180 days, every 5 days.
Moon series	Target Moon (301), center Earth (500@399), roughly ±15 days, every 12 hours.
Refresh behavior	Loaded when the scene is active in LIVE ORBITS mode; client refresh once per hour. API response cache up to one hour with stale-while-revalidate.

Scale and interpretation

The 3D renderer multiplies the Earth's heliocentric AU positions by 4.8 and the Moon's geocentric AU offsets by 520. It then adds the expanded lunar offset to the Earth's displayed position. These separate visual multipliers are deliberate so the lunar orbit remains visible; they mean the combined picture is not a single physically consistent scale model. Body radii

are also stylized. The timestamp and source vectors are data-driven, but visual distances and sizes must not be measured from the scene.

How the user can travel in the viewer

- Rotate: drag with the primary pointer or touch drag.
- Zoom: wheel/pinch, or the labeled zoom buttons.
- Pan: right-drag or two-finger pan; use Reset view to restore the standard camera.
- Inspect: select Moon, Sun, or Earth in the scene or body selector to reveal the matching source note.

Separate surface mode from macro-orbit mode

SITE VIEW places the selected site on the lunar globe and aims the camera toward the site. Its Sun/Earth directions follow either the shared SPICE cursor or LIVE NOW. LIVE ORBITS shows heliocentric/geocentric context and does not place the selected landing-site marker into that large-scale view.

Operational failure behavior

If a live endpoint fails, LIVE NOW and LIVE ORBITS display a committed NASA/JPL Horizons snapshot captured at the UTC time shown in the panel. Each fallback is labelled cached and is not presented as current telemetry; the reproducible SPICE dataset remains available for SITE VIEW and the explorer.

07 · ARCHITECTURE AND REPOSITORY

How the project is assembled

Layer	Responsibilities	Key files
Site and analysis config	Cited coordinate catalog, date window, step, frame, aberration, verdict thresholds.	config/sites.json; config/analysis.json; config/verdict.json
Scientific precompute	Load kernels, calculate site geometry and events, write provenance-rich JSON.	scripts/precompute.py; data/kernels/; data/kernel_manifest.json
Independent validation	Compare saved Horizons fixture samples against generated SPICE outputs.	scripts/validate_horizons.py; data/fixtures/; docs/validation.md
Static frontend data	Precomputed per-site time series and visual textures/video.	frontend/public/data/; frontend/public/textures/; frontend/public/videos/
React application	Narrative, explorer, compare, local sky, 3D viewer, audio, motion and state.	frontend/src/App.tsx; frontend/src/features/; frontend/src/state/
Live serverless API	Proxies selected Horizons observer and vector requests; Vite local middleware.	api/horizons.ts; api/orbits.ts; vercel.json

Frontend stack

The app uses TypeScript, React, Vite, Three.js with React Three Fiber, Zustand for explorer state, GSAP/ScrollTrigger and Lenis for the page narrative, custom CSS/Tailwind utilities, and custom SVG/canvas visualizations. The 3D scene is loaded lazily near the viewport. Reduced-motion and touch behavior are accounted for in the interaction design.

Key paths

- Project decisions and constraints: AGENTS.md.
- Scientific method and limitations: docs/scientific-method.md.
- Validation approach and results: docs/validation.md.
- Kernel download notes: data/README.md and fetch_data.sh.
- Frontend entry and user experience: frontend/src/App.tsx.
- Interactive 3D and orbit scene: frontend/src/features/OrbitalScene.tsx.
- Current JPL data clients and API routes: frontend/src/data/liveHorizons.ts, frontend/src/data/liveOrbits.ts, api/horizons.ts, api/orbits.ts.

Development setup

The repository documents verified commands for an environment where Python dependencies and kernels are already present. Fresh kernel downloads are documented by fetch_data.sh, whose own header notes the live-server download path was not tested in that checkout. A first-time developer can use:

```
python3 -m venv .venv . .venv/bin/activate pip install -r requirements.txt ./fetch_data.sh PYTHONPATH=scripts python scripts/precompute.py cd frontend npm install npm run dev
```

For the data currently included in the checkout, README.md lists `PYTHONPATH=scripts .venv/bin/python scripts/precompute.py`, the site-specific fixture validator, and frontend `npm run test` / `npm run build`. Before release, check that commands match the environment and that Vercel deployment has both API functions available.

08 · VALIDATION, LIMITS AND NEXT WORK

What has been checked

Saved Horizons observer-table fixtures check Sun (10) and Earth (399) azimuth/elevation at three epochs for all five sites in the 2027-01-01 through 2027-03-02 generated window. The tolerance is 0.1° for each angle.

Checked site	Samples	Largest reported difference
Apollo 11	3 epochs × Sun/Earth	Azimuth 0.004° · elevation 0.003°
Malapert Massif	3 epochs × Sun/Earth	Azimuth 0.005° · elevation 0.005°
Apollo 12	3 epochs × Sun/Earth	Azimuth 0.005° · elevation 0.004°
Shackleton Rim	3 epochs × Sun/Earth	Azimuth 0.005° · elevation 0.005°
de Gerlache Rim 1	3 epochs × Sun/Earth	Azimuth 0.003° · elevation 0.003°

Across all 30 observer comparisons, largest differences were 0.005° in azimuth and 0.005° in elevation. Six refined event checks at Apollo 11 and de Gerlache Rim 1 differed from Horizons geometric RAD events by at most 58.0 seconds (120-second tolerance). Horizons uses a target-semidiameter elevation cutoff to compare limb markers with LUMEN center crossings. DE440/DE441 and future UTC conversion are possible residual sources, not proven causes.

Known limits to state with results

- Terrain is not modeled. The apparent local horizon is a flat reference-sphere horizon, a major limitation for polar candidates.
- Current Horizons values are ephemeris snapshots, not live telemetry or measured lunar surface conditions.
- The orbit illustration expands distances and bodies; do not measure scale from its rendered spacing.
- Moon/Earth/Sun textures are visual reference assets and may not match the API epoch.
- The event checks cover selected crossings only; they do not independently validate every event or every time-series sample.
- The documented checks do not independently validate every time-series sample, refined event crossing, or landing suitability.
- The verdict thresholds are placeholders, not NASA criteria, and visibility does not establish communications availability.

Recommended next work

1. Complete clean-clone release verification, API hardening, and production smoke checks; record any external deployment blockers.
2. Run keyboard, reduced-motion, and mobile quality checks and record measured results.
3. Manually inspect narrow/mobile, keyboard-only, and reduced-motion flows, including live API failure and recovery states.
4. Keep a release checklist for actual deployment URL, same-origin `/api/horizons` and `/api/orbits`, and a private-window smoke check.
5. Only after v1 validation and release, consider terrain horizons from a selected NASA/PGDA LOLA product with documented raster frame, registration, uncertainty, and sensitivity checks.

Release statement

A strong LUMEN demo shows where a number came from, which time it describes, which horizon model was used, and which independent checks support it. The current result is geometric screening only.

09 · SOURCES AND GLOSSARY

Primary references

The following source links are official agency or project references used by the implementation or its method notes. The workspace keeps coordinates, kernel names, and validation fixtures locally so a user can inspect their provenance.

Reference	Use
JPL Horizons API documentation	API parameters and ephemeris/vector interfaces.
JPL Horizons manual	Observer and vector ephemeris usage reference.
NAIF generic kernels	SPICE kernel source and distribution.
NASA Apollo 11 landing coordinates	Reference site provenance.
NASA/NSSDC lunar landing sites	Apollo reference coordinates.
NASA Lunar Surface Data Book	South-pole candidate coordinate table.
NASA GSFC PGDA	South-pole data product context.
NASA GSFC SVS Moon visual source	LRO/LROC WAC visual texture source.
NASA Blue Marble	Earth visual texture source.
NASA SDO/AIA 171 Å image	False-color solar image used as a visual texture.

Glossary

Term	Meaning in this project
Azimuth	Horizontal direction measured clockwise from local north.
Elevation	Angle above or below the modeled local horizontal plane.
ENU	East-North-Up local coordinate basis at the lunar site.
ET / TDB	SPICE internal ephemeris time scale; UTC is used at the user-facing edges.
Horizons snapshot	A current ephemeris result for a UTC minute; not telemetry.
LT+S	SPICE light-time plus stellar-aberration correction used by the precomputed site geometry.
MOON_ME_DE440_ME421	Explicit lunar Mean Earth body-fixed frame selected in the project configuration.
SPICE	NASA/JPL toolkit and kernels used here to compute reproducible celestial geometry.
VEC_CORR=NONE	Horizons geometric vectors without light-time/aberration corrections.

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